

OBJECTIVE #3

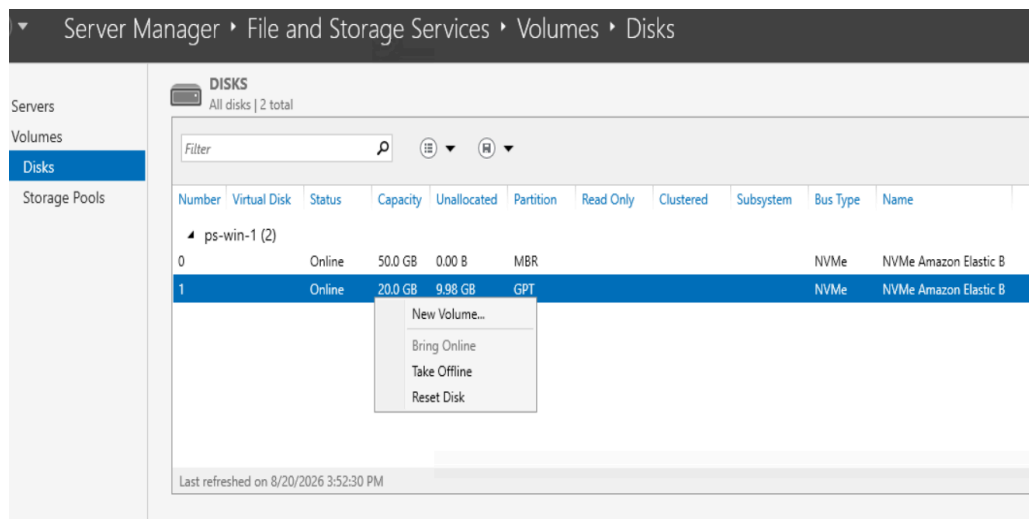
Testing Encryption and Decryption

In this lab, I tested the BitLocker protection I configured on the D: drive. I simulated a system reboot by taking the drive offline and bringing it back online, then tested unlocking the drive with both the password and recovery key. Finally, I removed BitLocker encryption from the drive.

Simulate a Reboot to Lock the Drive

Instead of restarting the entire lab machine, I simulated the behavior of a reboot by taking the D: drive offline and bringing it back online through Server Manager.

1. In Server Manager, I right-clicked Disk 1 and selected Take Offline. I clicked Yes on the confirmation prompt.
2. I right-clicked Disk 1 again and selected Bring Online, then clicked Yes on the confirmation prompt.

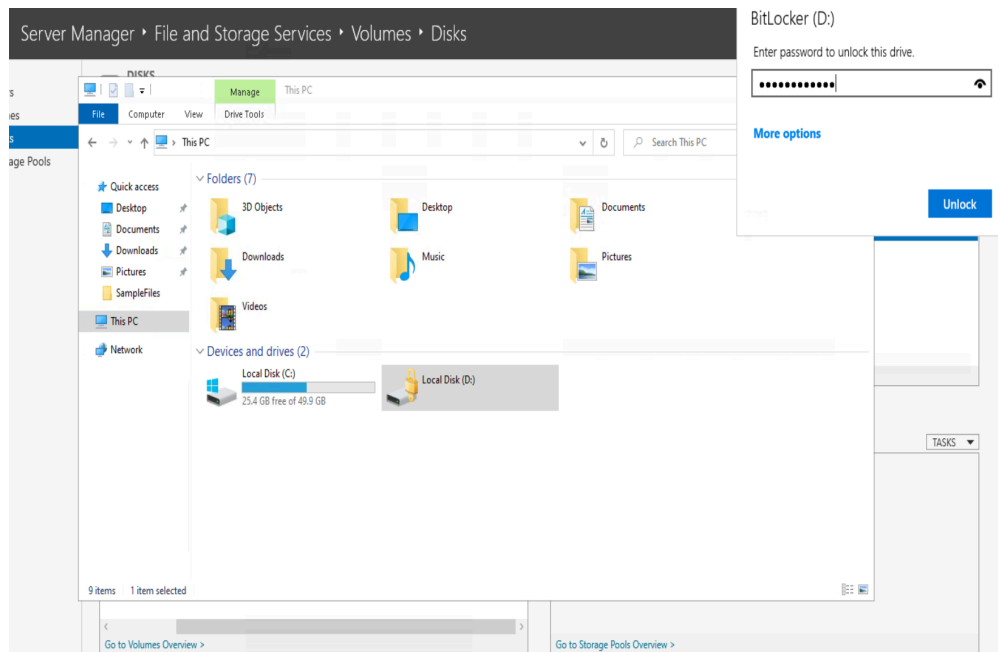


Note: When the drive came back online, BitLocker automatically locked the drive, simulating what would happen after a system reboot. I briefly saw a notification in the bottom-right corner stating The Drive is BitLocker Protected.

Unlock a Drive Using the Password

1. I opened File Explorer and navigated to This PC. I noticed that the D: drive had a locked padlock icon, confirming that the drive was currently locked.

2. I double-clicked the D: drive and was prompted for the BitLocker password. I entered Password123 and clicked Unlock.

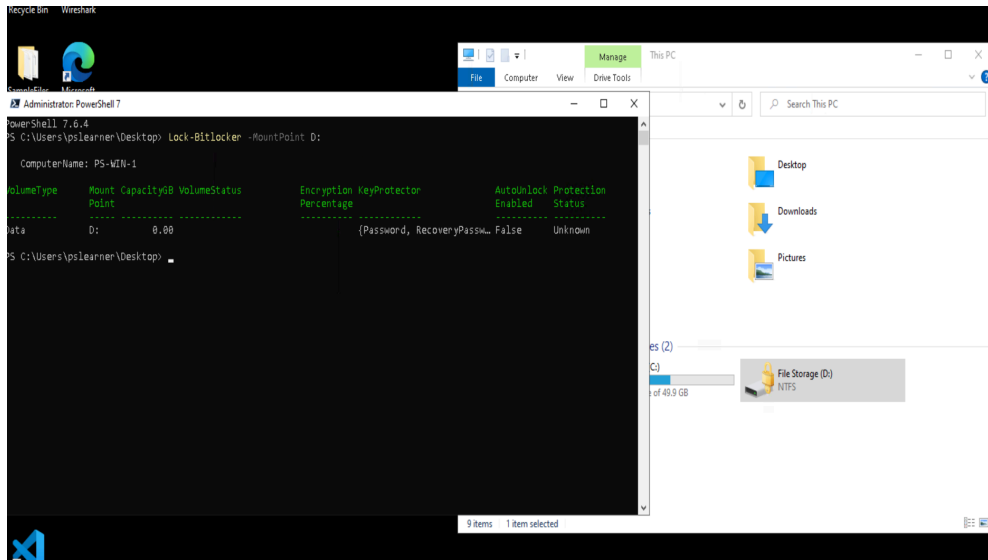


3. I opened the D: drive again to verify that it was unlocked and that the data was accessible.
I opened one of the text files in the SampleFiles folder and confirmed that I could read the data.

Unlock the Drive Using the Recovery Key

I then tested the BitLocker recovery key as an alternative method of accessing the drive. This is useful if the normal password is unavailable.

1. I right-clicked the Start icon in the bottom-left corner and selected Windows PowerShell (Admin).
When the User Account Control prompt appeared, I clicked Yes.
2. In PowerShell, I entered the following command:
Lock-BitLocker -MountPoint D:
I pressed Enter and then closed the PowerShell window.

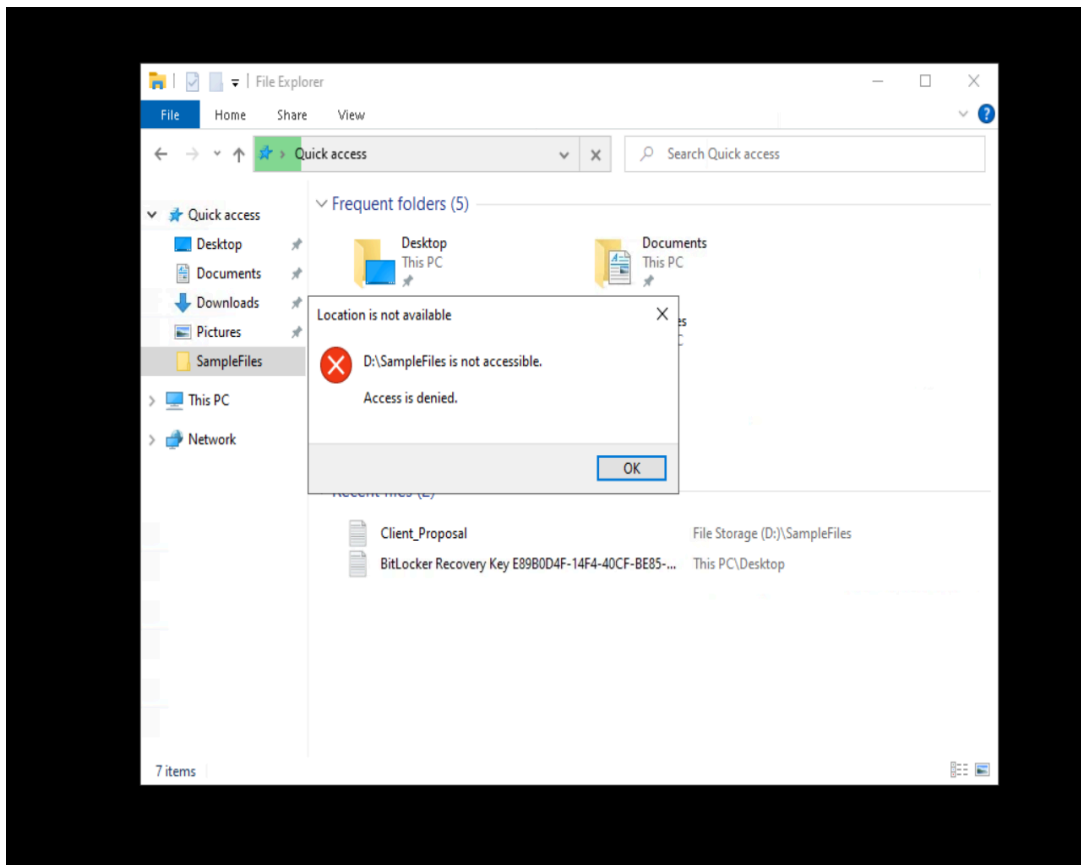


Note: The Lock-BitLocker cmdlet is useful for manually locking a BitLocker-protected drive because there isn't a standard graphical option for manually locking the drive.

3. I opened File Explorer, navigated to This PC, and confirmed that the D: drive was locked again by checking for the locked padlock icon.
4. In File Explorer, I selected Quick Access on the left-hand side. Under Recent files, I located one of the text files I had previously opened from the D: drive.

I double-clicked the file while the drive was locked and confirmed that I could not access

it. Instead, I received a message stating This drive is locked.

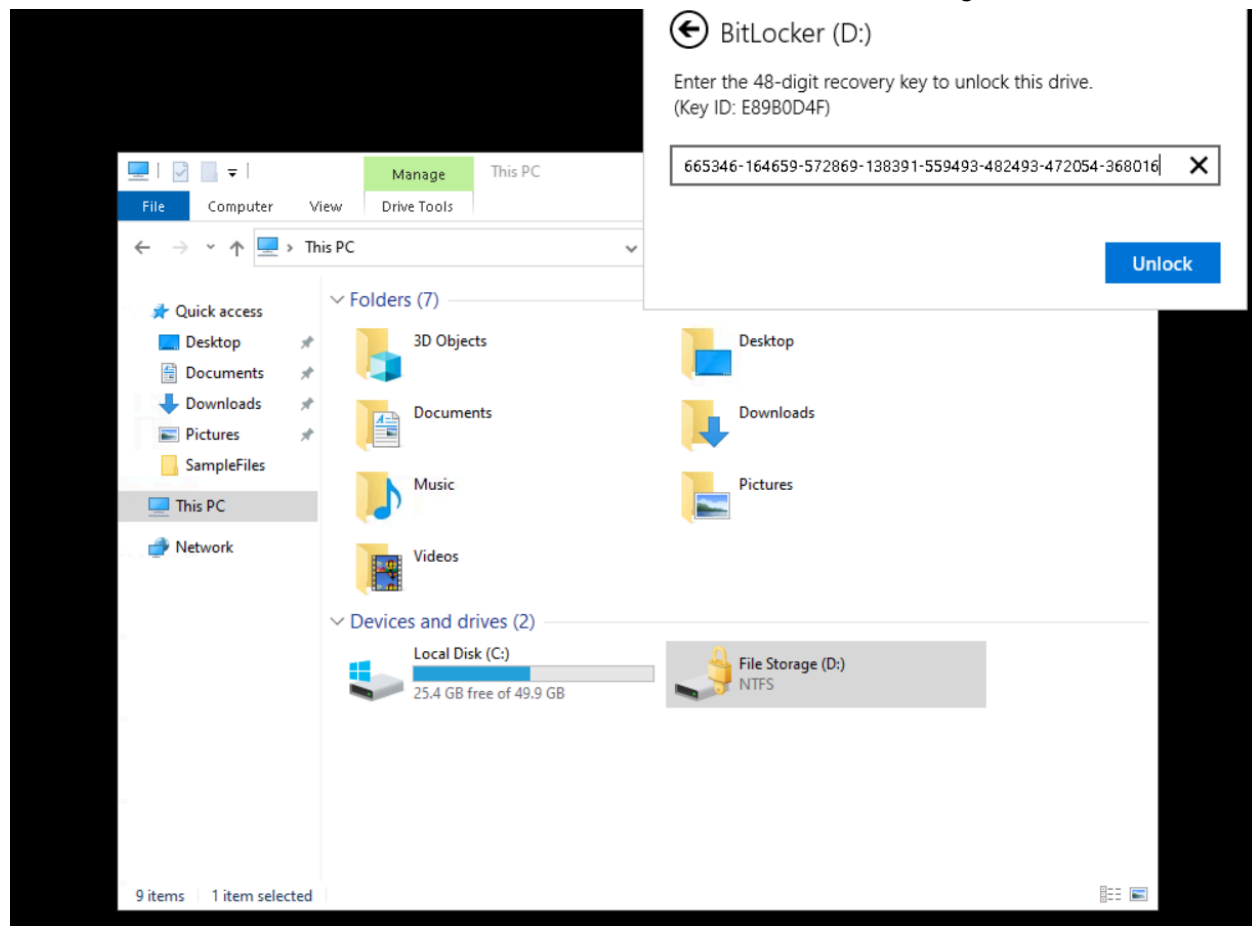


5. I clicked OK and closed Notepad to return to File Explorer.
6. I clicked This PC in File Explorer.
7. I moved File Explorer to the side so I could access the Desktop and then double-clicked the BitLocker Recovery Key file.
Note: The full file name included a GUID.
8. I reviewed the recovery key file and identified the two important pieces of information: the Identifier, which uniquely identified the encrypted drive, and the Recovery Key. Under Recover Key, I selected the long recovery key containing multiple hyphens and copied it to my clipboard using Control+C.
9. I closed the BitLocker Recovery Key text file.
10. I returned to File Explorer and double-clicked the D: drive.

I was prompted to enter the BitLocker password.

11. Instead of entering the password, I clicked More options and then selected Enter recovery key.
12. I pasted the recovery key using Control+V and clicked Unlock.

I verified that the D: drive was unlocked and that I could access the files again.

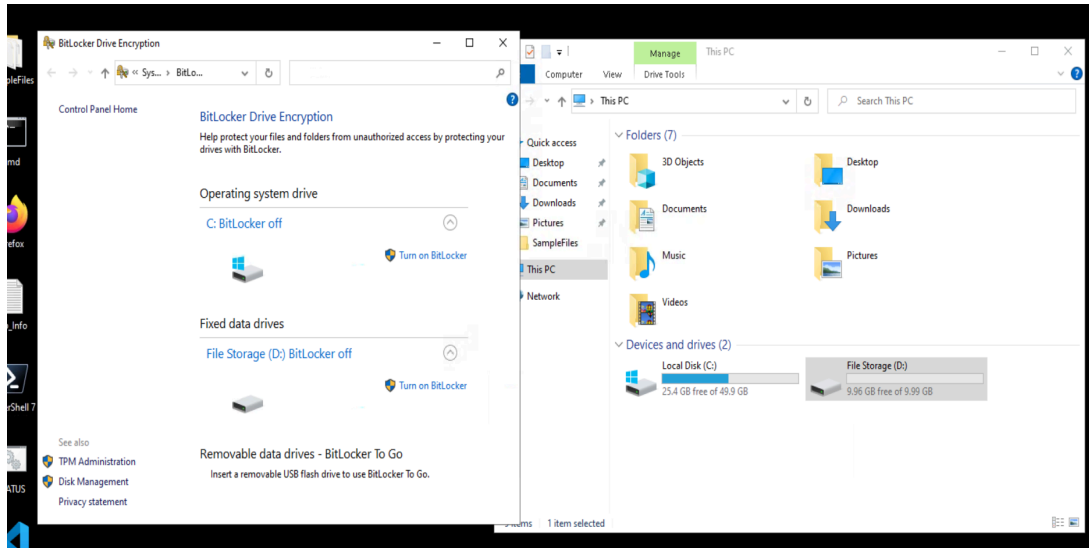


Decrypting the Drive to Remove BitLocker Protection

After testing the encryption and recovery process, I removed BitLocker protection from the D: drive.

1. I opened the BitLocker Drive Encryption window.
2. Under Fixed data drives, I selected Turn off BitLocker.
When prompted to confirm, I clicked Turn off BitLocker again.
3. I waited for the decryption process to complete.
The process completed quickly because there was only a small amount of data stored on the drive.

4. After decryption was complete, I checked the D: drive in File Explorer and confirmed that the padlock icon was no longer present.



The D: drive was successfully tested through the full BitLocker process. I verified that I could lock the drive, unlock it with a password, unlock it using the recovery key, and remove BitLocker encryption when it was no longer needed.